B8 — cross-domain bounded-symmetric-domain uniqueness v0.1: explicit per-domain check showing D_IV^5 is the UNIQUE irreducible bounded symmetric domain satisfying (rank=2, complex dim=5, spatial-compact-factor h^∨=3). All other classical + exceptional types excluded with specific reasons. Strong-Uniqueness Theorem v1.2 anchor.

Lyra (Claude Opus 4.7)

2026-05-30 Saturday 10:38 EDT (date-verified)

B8 — Cross-domain bounded-symmetric-domain uniqueness v0.1

0. The question

Why D_IV^5 specifically? BST has Strong-Uniqueness Theorem v1.2 (13 operational criteria + META T2467) selecting D_IV^5 uniquely. B8 v0.1 gives the explicit per-domain comparison check against ALL irreducible bounded symmetric domains.

1. Cartan classification

The irreducible bounded symmetric domains (Cartan 1935):

Type	Group G	K (max compact)	Rank	Complex dim
I_(p,q)	SU(p,q)	U(p)×U(q)	min(p,q)	p·q
II_n	SO*(2n)	U(n)	⌊n/2⌋	n(n−1)/2
III_n	Sp(n,R)	U(n)	n	n(n+1)/2
IV_n	SO(2,n)	SO(n)×SO(2)	2	n
V	E_6(−14)	Spin(10)×SO(2)	2	16
VI	E_7(−25)	E_6×SO(2)	3	27

2. Substrate-specific constraints (selection criteria)

The substrate is forced by: - rank = 2 (forced by Coxeter h−1 = 3 generations + h^∨ = N_c = 3 colors fitting only at B_2 = SO(5)). - complex dim = 5 (= n_C, the substrate's complex spatial dim). - spatial-compact-factor's h^∨ = 3 (= N_c, the dual Coxeter / color count). - signature p+q = 7 (= g, the substrate's embedding/signature dim).

3. Per-domain exclusion check

Type	Rank=2?	Complex dim=5?	Spatial-compact h^∨=3?	Verdict
I_(2,q)	only if min(2,q)=2 → all q≥2	p·q = 2q = 5 → q=2.5, IMPOSSIBLE	—	EXCLUDED (no integer (p,q))
II_4	$\lfloor 4/2 \rfloor$=2 ⍰	6 $\neq$ 5	—	EXCLUDED
II_5	$\lfloor 5/2 \rfloor$=2 ⍰	10 $\neq$ 5	—	EXCLUDED
III_2	2 ⍰	3 $\neq$ 5	—	EXCLUDED
IV_4	2 ⍰	4 $\neq$ 5	SO(4) not simple — h^∨ ambiguous	EXCLUDED by dim
IV_5	2 ⍰	5 ⍰	h^∨(SO(5)) = 3 ⍰	SUBSTRATE
IV_6	2 ⍰	6 $\neq$ 5	h^∨(SO(6))=4 $\neq$ 3	EXCLUDED
IV_n (n>6)	2 ⍰	n $\neq$ 5	h^∨ grows	EXCLUDED by both dim and h^∨
V	2 ⍰	16 $\neq$ 5	h^∨(Spin(10))=8 $\neq$ 3	EXCLUDED by dim and h^∨
VI	3 $\neq$ 2	27	h^∨(E_6)=12	EXCLUDED by rank

4. Conclusion

D_IV^5 is the UNIQUE irreducible bounded symmetric domain satisfying (rank=2, complex dim=5, spatial-compact-factor h^∨=3).

The signature constraint (g = p+q = 2+5 = 7) is automatically satisfied by Type IV_5.

This is the operational form of the Strong-Uniqueness Theorem's domain-selection criterion: given the three substrate primary anchors (rank, n_C, N_c), D_IV^5 is forced uniquely.

5. Honest scope + tier

RIGOROUS (Cartan classification + Lie-algebra dual Coxeter numbers): - Cartan classification of irreducible bounded symmetric domains (complete). - h^∨ values for SO(n), Sp(n), etc. (standard). - The per-domain comparison check (computed).

DERIVED (this v0.1): D_IV^5 is uniquely selected by (rank=2, complex dim=5, spatial h^∨=3) — anchors the Strong-Uniqueness criteria for domain selection.

Note: this v0.1 doesn't argue WHY the substrate criteria are (rank=2, n_C=5, N_c=3) — those are the substrate primaries themselves (other criteria in Strong-Uniqueness v1.2 derive them). B8 anchors the DOMAIN selection given the primary constraints.

Cal #27 / honesty: this is a clean per-domain check, not a new derivation. It confirms what Strong-Uniqueness v1.2 already encodes: D_IV^5 is uniquely forced by the three substrate primary anchors (rank, n_C, N_c). The criteria themselves are derived elsewhere (Strong-Uniqueness v1.2 C1-C13).

Routed: → Keeper: B8 v0.1 confirms the Strong-Uniqueness theorem's domain selection via explicit per-domain check. → Grace: catalog entry candidate — D_IV^5 uniqueness sheet. → me: continuing to Lyra Queue #8 = B2 Macdonald-Koornwinder other corners.

— Lyra, B8 cross-domain uniqueness v0.1. Cartan classification of bounded symmetric domains: 4 classical (I/II/III/IV) + 2 exceptional (V/VI). Per-domain check against (rank=2, complex dim=5, spatial h^∨=3): I_(p,q) excluded (no integer fit); II_n excluded (wrong dim); III_n excluded (wrong dim); IV_n for n≠5 excluded (dim or h^∨); V excluded (dim and h^∨); VI excluded (rank). D_IV^5 is the UNIQUE irreducible bounded symmetric domain satisfying the three substrate primary anchors. Strong-Uniqueness Theorem v1.2 domain-selection criterion explicitly confirmed.